

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper: BCS-504A — System Analysis Design

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER ANY FIVE QUESTIONS AND THE FIGURES IN THE RIGHT HAND MARGIN INDICATE:

Question 1.

- (a) Write the IEEE definitions of the following terms in connection with software: [6]
- (i) Error
 - (ii) Fault
 - (iii) Failure
- (b) Design an ER diagram for a Library Management System. Specify all constraints that should hold on the database. Make sure that it has at least three entity types, two relationship types, a weak entity type, a superclass/subclass relationship, and an n -ary ($n > 2$) relationship type. [8]

Question 2.

- (a) Describe software crisis and the goals of software engineering. [7]
- (b) Write four important attributes of software products and differentiate them from hardware products. [7]

Question 3.

- (a) Compare prototyping and iterative software development process models by listing their strengths and weaknesses. [7]
- (b) Describe at least five important characteristics of an SRS. [7]

Question 4.

- (a) What is the role of problem analysis in software development? Explain the use of Data Flow Diagrams in problem analysis with an example. [7]
- (b) What are use cases? Where are these use cases used during the software development process? Explain through an example. [7]

Question 5.

- (a) What do you understand by a module? What are the modularization criteria used for function-oriented design to produce modular designs? [7]

- (b) Explain all four major steps of structured design methodology using one example. [7]

Question 6.

- (a) Write the purpose of the following diagrams of UML used in object-oriented design: [6]

- (i) Class diagram
- (ii) Interaction diagram
- (iii) Activity diagram

- (b) What do you understand by test cases and test criteria? Write the differences between black-box and white-box testing. [8]

Question 7. Write short notes on the following: [14]

- (a) Test oracle
- (b) Software maintenance
- (c) Data dictionary
- (d) Open-closed principle